

CSC446 Project*

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1 Project Description

For the course project, I will be implementing Finite Element Method in 3D, using isoparametric quadratic elements, for the Laplacian PDE: $-\nabla^2 u(\vec{x}) = 0$. I will be using quadratic tetrahedral Lagrange basis, and also discuss mesh generation under *gmsh*, exploring three case studies. I will further be using Dirichlet Boundary Condition, based on the harmonic function $g(\vec{x}) = -\log(\|\vec{x} - \langle -7.1, -7.2, -7.3 \rangle\|)$.

The case studies are as follows:

1. Radius 5 sphere centered at the origin (Figure 1).

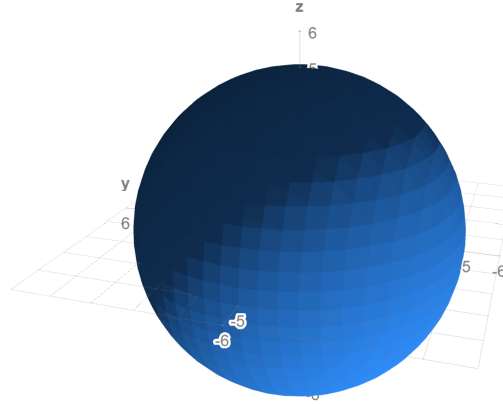


Figure 1: Radius 5 sphere.

2. Cylinder around the z-axis with radius 3 between $z = -5$ and $z = 5$ (Figure 2).
3. Cone between $z = 0$ and $z = 1$ with apex at origin and internal angle 5° (Figure 3).

2 Quadratic Tetrahedral Lagrange Basis

When considering 3D elements, the simplest choices include tetrahedrons and hexahedrons [1]. We will be using the unit tetrahedron, and transform it to physical element under a bijection to the same basis (isoparametric), and use these elements to fill our domain.

Table 1 lays out the quadratic tetrahedral lagrange basis in (ξ_1, ξ_2, ξ_3) space.

*[Project Code Link](#)

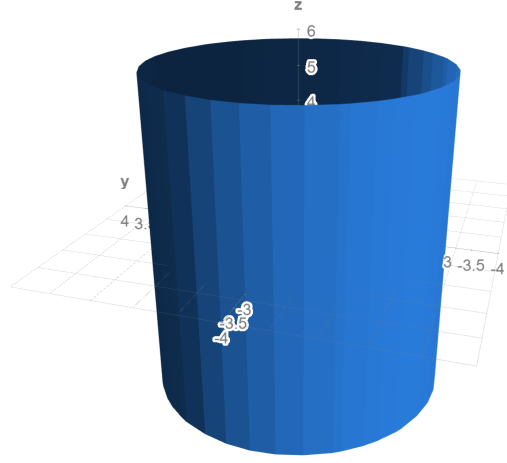


Figure 2: Radius 3 cylinder with height 10.

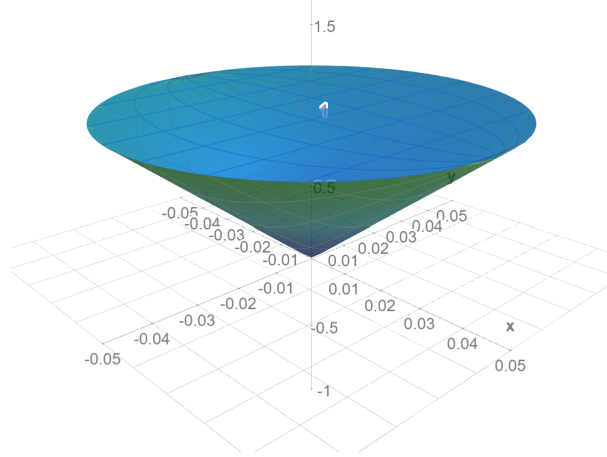


Figure 3: Sharp Cone of height 1, top radius $\tan(2.5^\circ)$.

Node Point z_i	Shape Function ϕ_i
$z_1(1, 0, 0)$	$\phi_1 = \xi_1(2\xi_1 - 1)$
$z_2(0, 1, 0)$	$\phi_2 = \xi_2(2\xi_2 - 1)$
$z_3(0, 0, 0)$	$\phi_3 = (1 - \xi_1 - \xi_2 - \xi_3)(1 - 2\xi_1 - 2\xi_2 - 2\xi_3)$
$z_4(0, 0, 1)$	$\phi_4 = \xi_3(2\xi_3 - 1)$
$z_5(\frac{1}{2}, \frac{1}{2}, 0)$	$\phi_5 = 4\xi_1\xi_2$
$z_6(0, \frac{1}{2}, 0)$	$\phi_6 = 4\xi_2(1 - \xi_1 - \xi_2 - \xi_3)$
$z_7(\frac{1}{2}, 0, 0)$	$\phi_7 = 4\xi_1(1 - \xi_1 - \xi_2 - \xi_3)$
$z_8(\frac{1}{2}, 0, \frac{1}{2})$	$\phi_8 = 4\xi_1\xi_3$
$z_9(0, \frac{1}{2}, \frac{1}{2})$	$\phi_9 = 4\xi_2\xi_3$
$z_{10}(0, 0, \frac{1}{2})$	$\phi_{10} = 4\xi_3(1 - \xi_1 - \xi_2 - \xi_3)$

Table 1: Node points and corresponding shape functions

Let the unit tetrahedron be $\mathbf{K}$ (Figure 4), and the bijective mapping from the finite element space $(\mathbf{K}, \mathcal{P}_2, \mathcal{N})$ be $F : \mathbf{K} \rightarrow \mathbb{R}^3$, mapping z_j to corresponding z_j^e in physical domain. Here, $\mathcal{N} = \{N_1, N_2, \dots, N_{10}\}$, with $N_j(v) = v(z_j)$. Then, the element basis functions over curved element $\mathbf{K}^e = F(\mathbf{K})$ is given by $\phi_j(F^{-1}(\vec{x}))$, and let's denote the isoparametric finite element space $(\mathbf{K}^e, \mathcal{P}_2^e, \mathcal{N}^e)$.

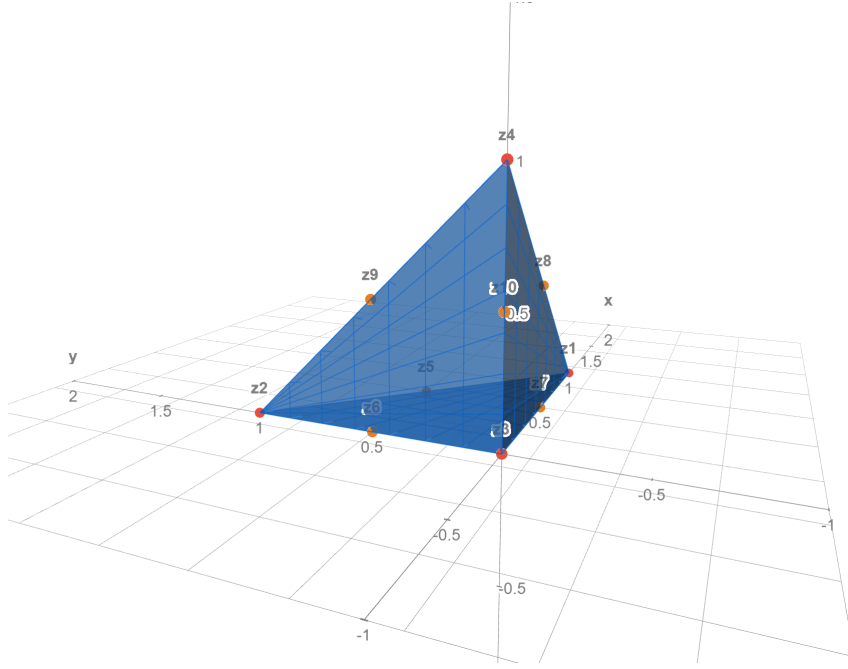


Figure 4: Regular Tetrahedron.

3 Weak Formulation & Stiffness Matrix

As seen in Homework 4, we will be relaxing the following weak formulation:

$$\int_{\Omega} \nabla u \cdot \nabla v \, dA = \int_{\partial\Omega} \frac{\partial u}{\partial n} v \, d\ell$$

If $\gamma u = g$, note that we are considering test functions over $V = \{v \in H^1\Omega \mid \gamma v = g\}$. Therefore, we will apply BC constraint to the linear system for the following problem explicitly:

$$\int_{\Omega} \nabla u \cdot \nabla v \, dA = 0$$

Equivalently, we will assemble the stiffness matrix $\mathbf{M}$ to solve for the vector of nodal values u take, $\mathbf{U}$, with rhs as the 0 vector, right before modifying this system with Dirichlet BC values by fixing node values at the boundary appropriately (substituting value in other rows as required). To assemble $\mathbf{M}$, we need the stiffness matrix over each element. When approximating with finite dimensional basis:

$$\mathbf{M}_{ij}^e = \int_{\mathbf{K}^e} \nabla_{\mathbf{x}} \phi_i(F^{-1}(\mathbf{x})) \cdot \nabla_{\mathbf{x}} \phi_j(F^{-1}(\mathbf{x})) \, d\mathbf{x}$$

Since $(x, y, z) = F(\xi_1, \xi_2, \xi_3)$, the Jacobian $J : \mathbf{K} \rightarrow \mathbb{R}^{3 \times 3}$ of F is:

$$J(\xi_1, \xi_2, \xi_3) = \begin{pmatrix} \frac{\partial x}{\partial \xi_1} & \frac{\partial y}{\partial \xi_1} & \frac{\partial z}{\partial \xi_1} \\ \frac{\partial x}{\partial \xi_2} & \frac{\partial y}{\partial \xi_2} & \frac{\partial z}{\partial \xi_2} \\ \frac{\partial x}{\partial \xi_3} & \frac{\partial y}{\partial \xi_3} & \frac{\partial z}{\partial \xi_3} \end{pmatrix}$$

Thus, substituting in our integral:

$$M_{ij}^e = \int_{\mathbf{K}} \nabla_{\mathbf{x}} \phi_i(\mathbf{t}) \cdot \nabla_{\mathbf{x}} \phi_j(\mathbf{t}) |J(\mathbf{t})| d\mathbf{t},$$

where $\mathbf{t} = (\xi_1, \xi_2, \xi_3)$, and $\nabla_{\mathbf{x}} \phi_i(\mathbf{t})$ is evaluated by:

$$\begin{pmatrix} \frac{\partial}{\partial x} \phi_i \\ \frac{\partial}{\partial y} \phi_i \\ \frac{\partial}{\partial z} \phi_i \end{pmatrix} = J^{-1}(\xi_1, \xi_2, \xi_3) \begin{pmatrix} \frac{\partial}{\partial \xi_1} \phi_i \\ \frac{\partial}{\partial \xi_2} \phi_i \\ \frac{\partial}{\partial \xi_3} \phi_i \end{pmatrix}.$$

We will be evaluating the integral using degree 2 quadrature rule [2] for $\mathcal{O}(h^3)$ accuracy, where h is the diameter of the tetrahedron:

$$\int_{\mathbf{K}} f(\mathbf{t}) d\mathbf{t} \approx \frac{1}{24} [f(b, b, b) + f(a, b, b) + f(b, a, b) + f(b, b, a)], a = 0.58541020, b = 0.13819660.$$

4 Mesh Generation

gmsh has a rich API to create and optimize meshes for objects in 1 to 3 dimensional spaces. We select Frontal-Delaunay mesh algorithm for surfaces in our 3D objects, and Frontal 3D mesh algorithm, running on top of OpenCASCADE CAD kernel. Frontal is necessary to generate quadratic order tetrahedral mesh for our volumetric objects. We further generate meshes for MSFs from 1/2 to 1/8. Figures 5, 6 & 7 provide a view of generated mesh for $MSF = 0.5$.

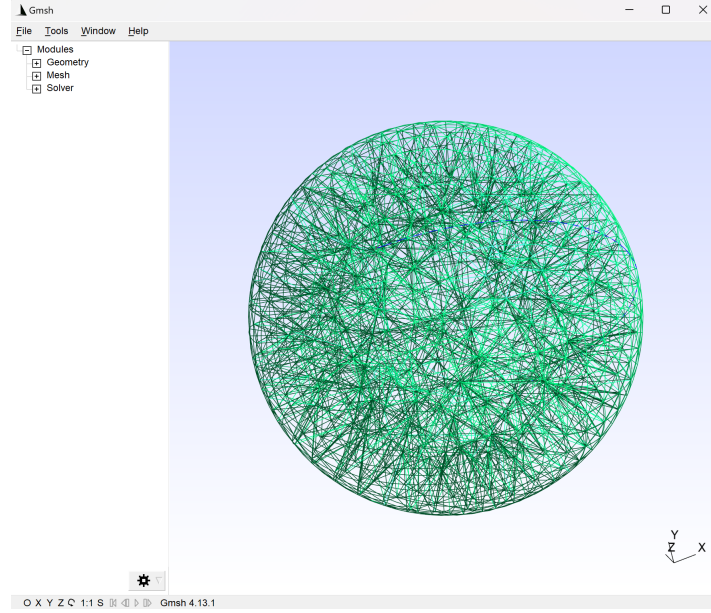


Figure 5: Mesh for sphere from case 1, $MSF = 0.5$.

gmsh also includes the ability to smooth corners, having an option which stores the number of mesh smoothing iterations. This can be particularly helpful when things misbehave around the boundary, specifically edges that don't form 90 degrees, causing singularities in solution. Thus, for case 3, given the apex of a sharp cone, we test the effect of smoothing by comparing unsmoothed mesh based solution and 10 iterations worth of smoothing based mesh solution.

5 Results

We calculate computed solution errors by:

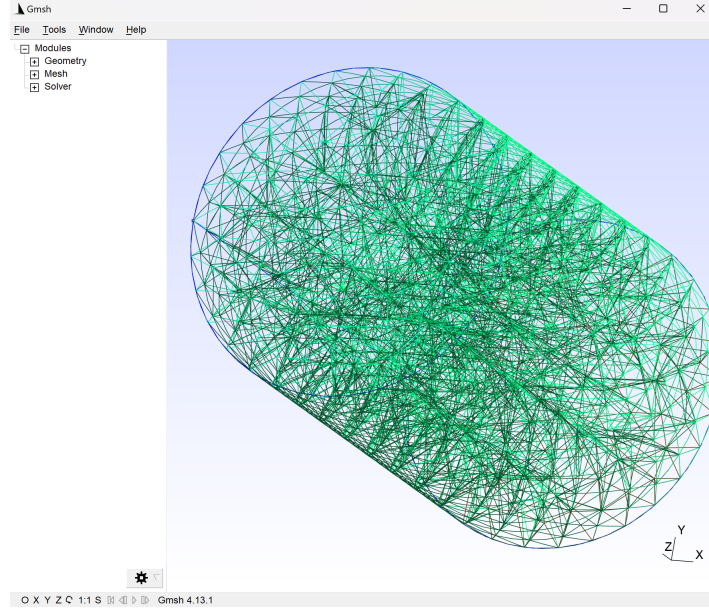


Figure 6: Mesh for cylinder from case 2, $MSF = 0.5$.

$$\varepsilon_u^h = \max_i |u_i^h - u(\mathbf{x}_i)|, \mathbf{x}_i \text{ node.}$$

Since g , the Dirichlet BC, is harmonic, $u = g$. Thus, we easily and quickly find max error using above formula. Below are observations for the different case studies:

- In case 1, we reach numerical limits on computation as we reach $MSF = 0.3$, performing worse than $MSF = 0.5$; The effect of computational limits is reached faster due to the pure curved surface of the case, a sphere, since generating mesh for curvature is generally much harder.
- We couldn't compute solution for $MSF = 1/8$ in case 2, given the large volume of the cylinder implying large number of finite elements. We observe relatively slower improvements with finer mesh otherwise, but not surprising given the majority curved surface of the cylinder.
- In case 3, we observe the benefits of smoothening, an improvement of $\approx 10^{-5}$. Note, we can potentially get worse solution if we over-smoothen sharp edges which are not 90 degrees!

Geometry	MSF	$h_{\max}$	ε_u^h
Case 1: Sphere	0.50	3.0870	0.0346
	0.35	1.9430	0.0331
	0.30	1.8449	0.0348
Case 2: Cylinder	0.50	1.9032	0.0219
	0.25	0.8747	0.0185
Case 3: Cone	0.29 (no-smooth)	0.0415	1.7204e-04
	0.29 (smooth)	0.0402	1.5950e-04

Table 2: Computed solution errors for cases 1,2 and 3.

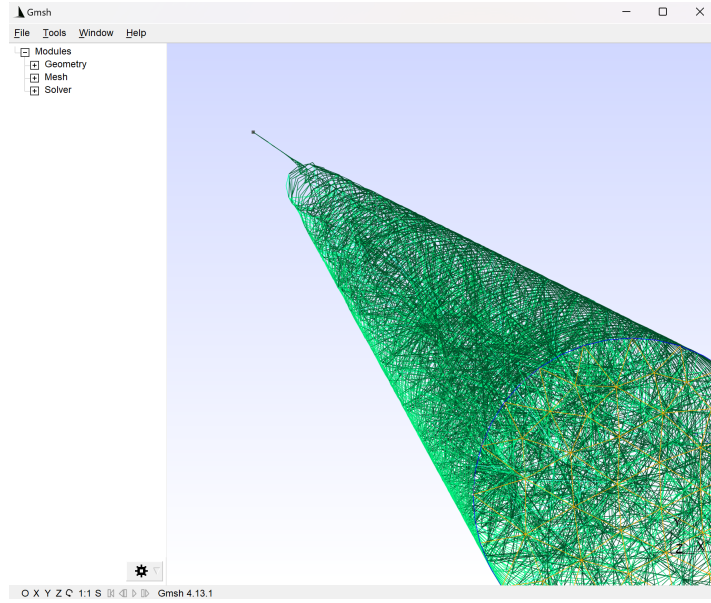


Figure 7: Mesh for cone from case 3, $MSF = 0.29$.

References

- [1] COSMOS. Shape function types (elements). https://doc.comsol.com/5.3/doc/com.comsol.help.comsol/comsol_api_xmesh.40.4.html.
- [2] Linbo Zhang, Tao Cui, and Hui Liu. A set of symmetric quadrature rules on triangles and tetrahedra. <https://lsec.cc.ac.cn/~tcui/myinfo/paper/quad.pdf>.